Abdoulaye Chaibou Saidou

Engineering student · Final-year internship — AI / Machine Learning Engineer

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PROFILE

Fifth-year engineering student (IATD-SI, ENSAM Meknès) seeking a final-year internship in AI/ML engineering. I have built and evaluated prediction models (LightGBM, Isolation Forest), recommenders and reinforcement-learning agents, and serve them through APIs (FastAPI, Docker). Internships in data science (2025) and full stack (2026).

Fast, continuous learner: picked up Julia, Rust, Unity / AR, RL and MLOps through my projects — I never stop learning.

SKILLS

ML: LightGBM, scikit-learn, Isolation Forest, SVM / decision trees, SVD / SVD++ / NCF, Soft Actor-Critic
Data: Python, pandas, NumPy, SQL (PostgreSQL), FFT, temporal and cross-validation
Deployment: FastAPI, Docker / Compose, Celery, Redis, TimescaleDB, Git
LLM: API integration (Groq, OpenRouter), validated structured outputs (Zod)
Other: TypeScript / Next.js, Julia, Rust, Java

EXPERIENCE

Full-Stack Developer Intern — DigiScia, Ouagadougou (Burkina Faso) Jul – Sep 2026

- SimpleOQuotidien project (connecting small-trade service providers with clients: search, quotes, assignment); my part: the front-end, in Next.js.

Data Scientist Intern (remote) — Technocolabs Softwares, Indore (India) Aug – Oct 2025

- Team project "Startup Acquisition Analysis & ML Pipeline": predicting startup outcomes (acquisition, closure, IPO, continued operation).
- My part: decision trees and SVM on the "companies" dataset.
- Contributed to preprocessing, feature engineering and cross-validation (per the certificate).

PROJECTS

Avilyre — supplier risk management SaaS Python, LightGBM, FastAPI, Next.js, Supabase

Academic capstone project, solo · avilyre.com

- Prediction service (LightGBM) for supplier delivery delay (classification) and PPM defect rate (regression).
- Leakage-safe features (expanding windows), temporal split, out-of-fold cross-validation, probability calibration; evaluated on synthetic data.
- LLM copilot (Groq / OpenRouter API) answering in natural language from the account's data; structured actions validated by schema and confirmed by the user.

AVIEN — industrial predictive maintenance FastAPI, Celery, Redis, TimescaleDB, Next.js, Docker Compose

Academic project, June 2026 · supervisor Prof. Smail Zaki

- Anomaly detection with Isolation Forest (100 trees, retrained every 24 h), weighted health score, linear remaining-useful-life (RUL) estimate with 95% confidence interval.
- FFT vibration diagnostics and occlusion-based explainability; limitations stated: simulated data, no quantitative evaluation.

Portefeuille-Advisor — reinforcement learning Gymnasium, stable-baselines3, Streamlit, TensorBoard

Academic project, Jan 2026 · supervisor Prof. Tawfik Masrour

- Gymnasium portfolio-management environment (3 assets, synthetic stochastic market, 120-month episodes) and Soft Actor-Critic agent trained for 3 risk profiles.
- Evaluated over 20 episodes against 3 benchmarks; Streamlit demo app.

Movie recommender system (MovieLens-1M) Python, scikit-learn, FastAPI, Streamlit

Group project (4 students), personal implementation

- Implemented collaborative-filtering, content-based, hybrid and advanced models (SVD, SVD++, NCF, Factorization Machines, ensemble) on 1 million ratings.
- Evaluation module (RMSE, MAE, Precision@K, NDCG@K), FastAPI API and Streamlit interface.

OrlVoice — voice assistant to control a computer Python, PyTorch, CamemBERT

Team project (OrlCheetah repository)

- Pipeline from speech recognition → NLP → action execution (open an application, click, type text); fine-tuned CamemBERT intent classifier.

EDUCATION

Engineering degree — ENSAM Meknès, IATD-SI track 2022 – 2027 (degree expected 2027)

LANGUAGES

French: working language, fluent · English: fluent written, intermediate spoken